

ZS-M43

The Z-Goldstone Is a Coherent Superfluid, and Its Dissipation Is Scrambling

Why the Effective-Hydrodynamics η/s Programme (Route B of ZS-M42) Dissolves, and the A-Scaled Unitary-Scrambling Structure That Replaces It — with an Anosov / Selberg Geometric Model and the $s = 1/2$ Confluence

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Verification: 22/22 structural-consistency PASS | Zero New Free Parameters | $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$ LOCKED. The Z-Goldstone is a coherent Euler/superfluid field (DERIVED-interpretation; corpus static instance $\rho_0 \propto 1/r^2$, DERIVED). It has no thermal viscosity: a thermal η/s and the KSS bound do not apply, because θ never thermalizes (ZS-F1 §7.2) and the dynamics is unitary (ZS-F1 §12.1). The correct, parameter-free dissipation is unitary scrambling, set by the i-tetration leak $1-|\lambda^2| \approx 0.205$ per cycle ($|\lambda^2| \approx 0.795$, PROVEN) and the master-equation rates $\lambda_{\text{fast}} = -A$, $\gamma_{xz} = 2A/Q$, $\gamma_{zy} = 6A/Q$ (PROVEN/DERIVED), propagating at a butterfly velocity $v_B \leq \rho(\mathcal{L}) \cdot a$ (ZS-M17). Its geometric model is Anosov geodesic mixing; via the Selberg bridge this meets ZS-M4's critical line at the $s = 1/2 \leftrightarrow \lambda = 1/4$ confluence (HYPOTHESIS-strong).

§0. Abstract

ZS-M42 [1] proved a Two-Gate No-Go for deriving the Navier–Stokes equations in Z-Spin and proposed, as Route B, an effective hydrodynamics built from the Goldstone phase θ of the Z-bias field. This paper resolves Route B. The Goldstone sector does furnish a coherent relativistic Euler / superfluid hydrodynamics, whose worked static instance is the corpus halo $\rho_0 \propto 1/r^2$ (ZS-A1/A2/F1 [3–5]). Its equation of state is fixed with no free parameter: a stiff fluid ($w = 1$, luminal sound speed $c_s = 1$) when the Goldstone evolves in time, and an anisotropic global-monopole configuration ($p_r = \rho$, $p_t = -\rho$, mean $\bar{w} = -1/3$) when it is the static halo. But the dissipative extension that Route B sought — an effective shear viscosity η , an entropy density s , and the ratio η/s — does not exist as a thermal object: θ never thermalizes (ZS-F1 §7.2 [5]) and the full dynamics is unitary, with scrambling rather than thermal entropy (ZS-F1 §12.1). The thermal-fluid framing fails a self-consistency test (a naive viscosity estimate is unstable and a thermal temperature is not uniquely fixed; §3 and Appendix C), which is the signature of a wrong identification, not of a hard calculation.

The positive result is that the correct dissipation is **unitary scrambling**, a parameter-free structure already present in the corpus: the i-tetration map at its fixed point z^* rotates by $\arg \lambda \approx 129.4^\circ$ per step and ‘leaks’ by $1-|\lambda^2| \approx 0.205$ per $X \rightarrow Z \rightarrow Y \rightarrow Z \rightarrow X$ cycle ($|\lambda^2| \approx 0.795$, ZS-M1 PROVEN), coarse-grained for X-observers into the master-equation rates $\lambda_{\text{fast}} = -A$, $\gamma_{xz} = 2A/Q$, $\gamma_{zy} = 6A/Q$ (ZS-Q7, ZS-F0). We quantify this with the out-of-time-order correlator (OTOC) and a butterfly velocity bounded by the Lieb–Robinson velocity $v_B \leq \rho(\mathcal{L}) \cdot a$ (ZS-M17 [2]); the Maldacena–Shenker–Starinets... [chaos] bound $\lambda_L \leq 2\pi T$ is noted to be inapplicable for the same non-thermal reason that retires η/s . Finally, the

i-tetration dilation rapidity $\alpha_{BK} = -\ln|z^*| = 0.5664$ (ZS-M4 Theorem 3, PROVEN) places the scrambling within the Berry–Keating / Anosov geodesic-flow picture; through the Selberg trace formula this meets ZS-M4’s critical line at the $s = 1/2 \leftrightarrow \lambda = 1/4$ confluence, where the 2025 Anantharaman–Monk optimal spectral gap [10,11] lives. We test whether the underlying operator map (ZS-M4’s transfer operator $\rightarrow$ a Selberg/Laplacian operator) can be built in the $P_{\max} \rightarrow \infty$ limit and find it blocked at the same gap that blocks ZS-M4’s Riemann connection, so the confluence remains HYPOTHESIS-strong (structural), not DERIVED. Several results are stated as formal claims: Lemma M43.1 (unitary + non-thermalizing $\Rightarrow \eta/s$ is the indeterminate 0/0, KSS vacuous); Proposition M43.2 (the X-observer entropy-production rate cross-checks the ZS-F16 Wilson dissipation to 0.16% and is bounded by the ZS-Q7 capacity $\ln 2$); and the identification of the stiff ($w = 1$) regime with the ZS-M12 centrifugal-launch term $Q_w^2/(2a^6\epsilon^2)$, $Q_w = A$, with the stiff field shown to be anisotropy-subdominant by a factor $\sim A^2$ near the cyclic bounce; and Lemma M43.3, a cardinality obstruction (an at-most-11-point spectrum cannot map to the infinite hyperbolic-Laplacian spectrum) that rigorously blocks the fixed-Q operator map. Zero new free parameters throughout.

Epistemic Status Legend

Status	Definition
PROVEN	Established with full rigor (here: cited from the corpus or external theorems).
DERIVED	Follows from the Z-Spin action + standard physics, zero free parameters.
DERIVED-interpretation	Synthetic reading of standard physics; not Z-Spin-specific.
HYPOTHESIS-strong	Multiple independent structural anchors; promotion path documented.
IMPORTED	Proved externally and used without re-proof; full citation given.
OPEN	Recognized gap honestly registered for future work.
NON-CLAIM	Explicit declaration of what is NOT asserted.

§1. Introduction

ZS-M42 [1] established that an ordinary micro-to-Navier–Stokes derivation in Z-Spin factors through a locality gate G1 (to which the ZS-M17 Lieb–Robinson estimate [2] contributes) and a chaos/entropy gate G2 (for which Z-Spin supplies nothing), and is therefore obstructed at G2. ZS-M42 §7 proposed Route B: build an effective hydrodynamics directly from the Goldstone phase θ . This paper carries Route B to its conclusion and finds a sharper, more useful result than the one sought. The Goldstone sector gives a coherent Euler/superfluid hydrodynamics (§2), but not a thermal one. §3 shows that the thermal viscosity / η/s programme is not merely hard but **ill-posed**: θ never thermalizes and the dynamics is unitary, so there is no thermal temperature and no thermal entropy, and the KSS bound does not apply. The constructive content begins in §4: the dissipation that physically exists in the Z-sector is **unitary scrambling**, a parameter-free structure. §5 quantifies it (OTOC, butterfly velocity), §6 gives its geometric model (Anosov mixing, the Selberg bridge, and the $s = 1/2$ confluence with ZS-M4, including a test of whether that confluence can be lifted to a theorem). §7 records gates and non-claims; §8 concludes.

§2. The Goldstone Phase as a Coherent Hydrodynamic Field

Write the Z-bias field as $\Phi = \rho e^{i\theta}$ with vacuum $|\Phi| = 1$; the radial mode ρ is heavy and the angular mode θ is an exactly massless Goldstone boson, with conserved winding charge $Q = a^3 \rho^2 \theta$ (ZS-F1 [5], ZS-U11 [6]). At the vacuum the phase-sector stress-energy is

$$T_{\mu\nu} = M_p^2 [\partial_\mu \theta \partial_\nu \theta - \frac{1}{2} g_{\mu\nu} (\partial\theta)^2] ,$$

and with $u_\mu \propto \partial_\mu \theta$, $n \propto \rho^2$, the conservation law $\partial^\mu T_{\mu\nu} = 0$ is the relativistic Euler equation of a perfect (dissipationless) superfluid — the standard Madelung / superfluid correspondence [12,13]. Its worked static, curl-free instance is in the corpus: the radial profile $\theta(r) = \ln(r/r_0)/L$ gives $\rho_0 \propto 1/r^2$, the isothermal halo of ZS-A1/A2 and ZS-F1 §4.3 [3–5]. The perfect-fluid content is standard physics; the Z-Spin-specific content is what plays the role of dissipation, to which the rest of the paper is devoted. [DERIVED-interpretation; static instance DERIVED in corpus.]

2.1 Equation of state and sound speed

The Euler system above has a definite equation of state, fixed with no free parameter by the canonical Lagrangian. In the homogeneous time-dependent (‘kination’) regime $\theta = \theta(t)$, the stress-energy gives $\rho = p = \frac{1}{2} M_p^2 \dot{\theta}^2$, so the equation-of-state parameter and sound speed are

$$w = p/\rho = 1 , \quad c_s^2 = dp/d\rho = 1 \Rightarrow c_s = 1 \quad (\text{luminal}).$$

The Z-Goldstone is thus a **stiff fluid** — the maximally stiff causal fluid — with sound speed equal to the speed of light; for any canonical $P(X) = X$ theory the general formula $c_s^2 = P_X/(P_X + 2X P_{XX})$ [21] reduces to 1. In the static, curl-free (halo) regime $\theta = \theta(r)$ with $\dot{\theta} = 0$ and only $\partial_r \theta \neq 0$, the stress-energy is instead anisotropic:

$$\rho = -T^t_t = \frac{1}{2} M_p^2 (\theta')^2 , \quad p_r = T^r_r = +\rho , \quad p_t = T^\theta_\theta = T^\phi_\phi = -\rho .$$

This is the global-monopole / texture gradient stress [22]: radial pressure equal to the energy density, tangential tension equal to minus it. The angle-averaged equation of state is $\bar{w} = (p_r + 2p_t)/3\rho = -1/3$, and since $\theta' \propto 1/r$ the density is $\rho \propto 1/r^2$ — recovering the corpus halo $\rho_0 \propto 1/r^2$ and identifying its effective equation of state as the coasting / flat-rotation value $\bar{w} = -1/3$ (the borderline between cosmic acceleration and deceleration, and the value that produces asymptotically flat rotation). Thus the same field is a stiff ($w = 1$) fluid when it evolves in time and a $\bar{w} = -1/3$ anisotropic configuration when it is the static halo — both with zero free parameters. [DERIVED.]

2.2 Cosmological placement of the stiff regime

The two equation-of-state regimes are not abstract: each is a known phase of the corpus cyclic cosmology, and the stiff regime in particular closes a connection that was previously left implicit. Writing the radial amplitude as $\varepsilon = |\Phi|$ (the ρ of §2, denoted ε in ZS-U5/M12/A2), the conserved comoving winding charge is $Q_w = a^3 \varepsilon^2 \theta$, which ZS-M12 / ZS-U5 fix at the Z-Telomere onset to $Q_w = \mathbf{A} = 35/437$ [25,26].

Eliminating $\theta = Q_w/(a^3 \varepsilon^2)$, the kinetic (stiff) energy density of the Goldstone is

$$\rho_0 = \frac{1}{2} \varepsilon^2 \theta^2 = Q_w^2 / (2 a^6 \varepsilon^2) \propto a^{-6} ,$$

which is exactly the centrifugal launch term $Q_w^2/(2a^6\epsilon^2)$ of the ZS-M12 ‘centrifugal launch mechanism’ [26]. The stiff $w = 1$ regime of §2.1 is therefore the **near-bounce phase** of the ZS-U11 cycle: as $a \rightarrow 0$ the conserved winding forces $\theta \propto a^{-3}$ and $\rho_\theta \propto a^{-6}$ to grow fastest, building the barrier $\epsilon_{\min}(\text{bounce}) \approx 30.7$ that places the field in the large-field regime required for inflation. The static $\bar{w} = -1/3$ configuration of §2.1 is the complementary late-time phase — the same Goldstone frozen into the halo gradient.

[DERIVED for the energy density; placement DERIVED-interpretation against ZS-M12/U11.]

Two honest qualifications. (i) Because $w = 1$ (not $w > 1$), $\rho_\theta \propto a^{-6}$ scales like the shear anisotropy rather than dominating it, so the Z-Goldstone is marginal for ekpyrotic smoothing [27]; the ZS-U11 bounce relies on the topological Auto-Surgery, not on ultra-stiff dilution. (ii) Whether Q_w survives the $\approx 3\tau_P$ dissipative Auto-Surgery is the standing ZS-M12 / ZS-U11 matching question (V1 OPEN, gate F-U11.6). We note — without claiming it — that this $\approx 3\tau_P$ duration is suggestively close to the ≈ 3 -cycle time for the Z-channel to saturate its $\ln 2$ capacity (§4.1); establishing or refuting that coincidence is left OPEN. [OPEN.]

The marginality in (i) can be made quantitative. In a Bianchi-I (anisotropic) background the shear energy density also scales as $\rho_{\text{shear}} = \sigma^2/16\pi G \propto a^{-6}$, identically to $\rho_\theta = Q_w^2/(2a^6\epsilon^2)$, so their ratio is a constant fixed by initial data: $\rho_\theta/\rho_{\text{shear}} = A^2/(2\epsilon^2 C_\sigma)$. With $Q_w = A$, the Goldstone coefficient is $A^2 \approx 6.4 \times 10^{-3}$; unless the shear is itself tuned below this, the Goldstone kination is anisotropy-**subdominant** by a factor $\sim A^2/2 \approx 0.003$ — it is neither catastrophically overwhelmed nor capable of smoothing, but it cannot be the smoothing agent (which would require $w > 1$). This is the deterministic complement to the corpus’s thermal-shear constraint $\tau_{\text{thermal}}/\tau_{\text{AS}} \approx 0.81$ (ZS-U11, channel P4, with $\tau_{\text{AS}} = 3\tau_P$), and reinforces the same conclusion: the bounce smoothing is supplied by the topological Auto-Surgery, not by stiff dilution. [DERIVED-CONDITIONAL on the shear not being tuned below A^2 .]

§3. There Is No Thermal Viscosity

A Navier–Stokes-like extension would need a shear viscosity η and, for the dimensionless ratio η/s , an entropy density s and a temperature T . Two standing corpus results block this. First, the Goldstone **never thermalizes**: at the attractor θ has zero direct coupling to matter (ZS-F1 §7.2 [5], gate F-F1.5) — it is a coherent classical field, the same one whose static profile gives $\rho_\theta \propto 1/r^2$, not a thermal gas. Second, the full Z-Spin dynamics is **unitary**, and the entropy seen by X-observers is scrambling/entanglement entropy, not thermal entropy (ZS-F1 §12.1). With no thermal T and no thermal s , the ratio η/s is not the right object: for a pure coherent superfluid the thermal normal component is absent, so a thermal η and s both vanish and η/s is $0/0$. The KSS bound $\eta/s \geq 1/4\pi$ [14], a theorem about thermal fluids, simply does not apply.

This is confirmed by a self-consistency test (detailed in Appendix C): every attempt to assign a viscosity value is unstable. Reading the Goldstone drag as a hydrodynamic mode-damping gives a small $\nu \sim A^2$, whose implied $\eta/s \approx 4 \times 10^{-4}$ lies $\sim 200\times$ below the KSS bound — unphysical for any thermal fluid; reading it as a microscopic relaxation rate gives a large $\nu \sim 1/A^2$; and the candidate temperatures (a cyclic-period value and a vortex-core value) differ by $\sim 2\times$, so no η/s value is parameter-free. An estimate that flips sign and magnitude with each re-derivation, while a required input (T) cannot be uniquely fixed, is the signature that the **imported framework is wrong** — here, that thermal hydrodynamics does not match the corpus’s unitary, non-thermalizing ontology. The instability is therefore informative, not a defect to be patched.

Lemma M43.1 (No-Thermal-Transport). Let a system evolve under a one-parameter unitary group $U(t) = e^{-iHt}$ (global unitarity), and suppose it does not thermalize: there is no inverse temperature β and Gibbs state $\rho_\beta = e^{-\beta H}/Z$ to which reduced observables relax (e.g. a conserved coherent mode with no relaxation channel, as for θ in ZS-F1 §7.2). Then: (a) the Kubo shear viscosity $\eta = \lim_{\omega \rightarrow 0} \omega^{-1} \text{Im } G^R(\omega, 0)$, defined through the KMS (thermal) stress-tensor correlator, has no thermal state in which to be evaluated; (b) the thermal entropy density $s = (\rho + p)/T$ is undefined, there being no T ; (c) in the pure-coherent (two-fluid $T \rightarrow 0$) limit the normal component that carries η and s vanishes, so $\eta \rightarrow 0$ and $s \rightarrow 0$ and the ratio η/s is the indeterminate form $0/0$. Hence the KSS bound $\eta/s \geq 1/4\pi$, a theorem about thermal fluids, constrains nothing here. The sole well-defined irreversibility is the entanglement-entropy production of a traced subsystem (§4.1), which is information-theoretic, not thermodynamic. [DERIVED — a careful application of the definitions (Kubo needs KMS; thermal s needs T) to the corpus’s unitarity and non-thermalization; not a new theorem.]

Remark (two-fluid limit). Part (c) is sharp in Landau’s two-fluid description [28]: the shear viscosity and the entropy reside entirely in the normal component of density $\rho_n(T)$ — $\eta \sim \rho_n \langle v \rangle \ell$ and $s \propto \rho_n$ — while the superfluid component ρ_s carries neither. As $T \rightarrow 0$ the thermal quasiparticles vanish, $\rho_n(T) \rightarrow 0$, so $\eta \rightarrow 0$ and $s \rightarrow 0$ together. The Z-Goldstone, a non-thermalizing coherent condensate, sits at $\rho_n \equiv 0$ (a $T = 0$ state), so $\eta = s = 0$ identically and $\eta/s = 0/0$. The KSS bound concerns the universal $T \rightarrow 0$ limit of strongly-coupled, normal-component-dominated plasmas; with no normal component, the Z-Goldstone lies outside its domain entirely. [DERIVED.]

§4. The Correct Dissipation: Z-Spin Scrambling

What physically dissipates in the Z-sector — the ‘leak’ an X-observer perceives — is unitary scrambling, and it is parameter-free. The i -tetration map $f(z) = i^z$ has the attracting fixed point $z^* = 0.4383 + 0.3606i$ with multiplier $\lambda = (i\pi/2)z^* = -0.5664 + 0.6886i$, $|\lambda| = 0.8916$ (ZS-M1 [8], PROVEN). Two PROVEN facts make it a scrambler rather than a simple sink:

(i) Quasi-periodic rotation. Because the real matrix M_f forces $|\lambda| = |\bar{\lambda}|$, no single ray is selected: the trajectory rotates by $\arg \lambda \approx 129.4^\circ$ per step within the 2-dimensional Z subspace (ZS-M1; ZS-F0 V35). This is the mechanism that, for X-observers, implements scrambling (ZS-F1 §12.1).

(ii) The Wilson-loop leak. The holonomy around one $X \rightarrow Z \rightarrow Y \rightarrow Z \rightarrow X$ cycle is $|\lambda^2| = (\pi^2/4) \cdot \eta_{\text{topo}} \approx 0.795$ (ZS-M1 Remark 1.2, PROVEN); the per-cycle leak is $1 - |\lambda^2| \approx 0.205$, equivalently a per-cycle entropy-production rate $-\ln|\lambda^2| \approx 0.229$. A closed loop ($|\lambda^2| = 1$) would be non-scrambling and marginally stable; the leak is precisely what scrambles.

Coarse-grained for an X-observer, this unitary scrambling is the Pauli/Lindblad master equation of ZS-Q7 [7], with the inter-sector rates

$$\lambda_{\text{fast}} = -\mathbf{A}, \quad \lambda_{\text{slow}} = -2\mathbf{A}/Q, \quad \gamma_{xz} = 2\mathbf{A}/Q, \quad \gamma_{zy} = 6\mathbf{A}/Q \quad (\text{ZS-Q7, ZS-F0; PROVEN/DERIVED}).$$

These are $\mathbf{A}$ -scaled and contain no free parameter. The decoherence time $\tau_{\text{fast}} = 1/\mathbf{A} \approx 12.49$ coincides with the Z-bottleneck relaxation of ZS-F0 and the decoherence ratio $\tau_D/\tau_{\text{Penrose}}$ of ZS-Q1. The scrambling, not a thermal viscosity, is the genuine Z-sector ‘dissipation’; it is information-theoretic and unitary at the fundamental level, dissipative only in the reduced X-description. [DERIVED/PROVEN.]

4.1 The dynamical character: scrambling without chaos

Precision about the dynamical type is essential, because it bounds what may be claimed. The i-tetration fixed point is **attracting**: $|\lambda| = 0.8916 < 1$, so $\ln|\lambda| = -0.115 < 0$. By Pesin's identity the Kolmogorov–Sinai entropy equals the sum of the positive Lyapunov exponents [23], and an attracting fixed point has none — so the map's KS entropy is zero and there is **no positive quantum Lyapunov exponent** λ_L . The system is therefore not chaotic in the dynamical-systems sense. ‘Scrambling’ here means unitary information delocalization, not chaotic-flow mixing: per $X \rightarrow Z \rightarrow Y \rightarrow Z \rightarrow X$ cycle a fraction $1 - |\lambda^2| \approx 0.205$ of the information leaks to the traced-out Z/Y sectors, giving the reduced X-observer an entropy growth $-\ln|\lambda^2| \approx 0.229$ per cycle — equivalently the CPTP master equation above. Two consequences follow. (i) The decoherence/scrambling time is a **fixed constant**, $\tau_{\text{fast}} = 1/A \approx 12.49$ (or $\approx 1/(-\ln|\lambda^2|) \approx 4.4$ Wilson cycles), independent of the system size S . (ii) The well-posed, parameter-free ‘irreversibility’ of the Z-sector is therefore not a transport coefficient η but the X-observer **entropy-production rate** $\dot{S}_X \approx (-\ln|\lambda^2|)/T_{\text{cycle}}$ — the Page-curve growth of ZS-F1 §12.1 — an information-theoretic quantity that is the correct replacement for the ill-posed η 's of §3. [DERIVED.]

Proposition M43.2 (cross-validation of $\dot{S}_X$). The rate $\dot{S}_X$ is consistent with two independent corpus quantities. (i) Exactly: per cycle, $\dot{S}_X \cdot T_{\text{cycle}} = -\ln|\lambda^2| = 2(-\ln|\lambda|) = 0.2294$, while the ZS-F16 Wilson-loop dissipation gives $\Gamma_Z \cdot T_{\text{cycle}} \approx 0.1149$; since $-\ln|\lambda| = 0.1147$, these agree to 0.16% — as they must, both being the same $|\lambda^2| = 0.7948$ Wilson loop, the factor 2 being the amplitude-to-probability relation. (ii) Approximately: the coarse-grained Markovian leak $\gamma_{XZ} \cdot \tau_{\text{fast}} = (2A/Q)(1/A) = 2/Q \approx 0.182$ reproduces the exact per-cycle linear leak $1 - |\lambda^2| \approx 0.205$ to $\sim 11\%$, the residual being the expected Markovian-approximation error (the rate misses the arg $\lambda \approx 129.4^\circ$ rotation). (iii) Bounded: the accumulated X-entropy saturates at the Z-channel capacity $\leq \ln 2$ (ZS-Q7) after $\approx \ln 2 / (-\ln|\lambda^2|) \approx 3.0$ cycles. Thus $\dot{S}_X$ is fixed (zero free parameters), cross-checked against ZS-F16 exactly and ZS-Q7 within the Markovian error, and capacity-bounded. [DERIVED; cross-checks consistent.]

§5. Scrambling Quantified: OTOC and the Butterfly Velocity

Scrambling is measured by the out-of-time-order correlator (OTOC) [15], $C(t) = \langle [W(t), V]^\dagger [W(t), V] \rangle$, whose growth defines a rate and a spatial spreading speed, the butterfly velocity v_B [16,17]. Two Z-Spin statements follow without new assumptions. The spreading speed is bounded by the Lieb–Robinson velocity of the Z-Spin lattice,

$$v_B \leq v_{LR} = \rho(\mathcal{L}) \cdot a \quad (\text{ZS-M17 [2], } \leq\text{-form per the 2026-06-07 erratum),}$$

and generically the inequality is strict (the butterfly/operator-growth velocity is typically below the Lieb–Robinson velocity), consistent with the erratum's downgrade of the M17.2 tightness claim. The per-cycle scrambling rate is the leak $-\ln|\lambda^2| \approx 0.229$ of §4.

One boundary must be respected. The Maldacena–Shenker–Stanford chaos bound $\lambda_L \leq 2\pi k_B T/\hbar$ [18] is a *thermal* bound; since the Z-Goldstone has no thermal temperature (§3), it does not apply here — exactly the inapplicability that retires η 's and the KSS bound. The scrambling is therefore characterized by its unitary, parameter-free leak and its sub-Lieb–Robinson light-cone, not by a thermal Lyapunov bound. [DERIVED-interpretation; external OTOC/butterfly framework IMPORTED.]

The non-chaotic character of §4.1 sharpens what the OTOC framework does and does not give. The butterfly **velocity** — the operator-spreading speed — is well-defined and bounded by $v_B \leq \rho(\mathcal{L}) \cdot a$ irrespective of chaos, because that bound is a Lieb–Robinson statement. But the exponential OTOC **growth** $e^{\{\lambda_L t\}}$ that marks a chaotic fast scrambler is absent here, since $\lambda_L = 0$ ($KS = 0$). In particular the fast-scrambling relation $t_* \sim (1/\lambda_L) \ln S$ [24] does **not** apply: the Z-sector is not a black-hole-like fast scrambler but a fixed-rate decoherer, with a size-independent scrambling time set by $1/A$ (§4.1). This is the OTOC counterpart of the MSS-bound inapplicability — both the chaos bound and the logarithmic fast-scrambling time presuppose the chaotic, thermal dynamics the Z-Goldstone does not have. [DERIVED.]

§6. The Geometric Model: Anosov Mixing, the Selberg Bridge, and the $s = 1/2$ Confluence

The scrambling has a natural geometric model, supplied by ZS-M4 [9]. Theorem 3 of ZS-M4 (PROVEN) identifies the Berry–Keating dilation rapidity with an i-tetration quantity,

$$\alpha_{BK} = \ln(\text{dilation}) = -\ln|z^*| = y^* \cdot \pi/2 = 0.566417 \quad (y^* = \text{Im } z^* = 0.36059; \text{locking L3: } |z^*|^2 = e^{\{-y^* \pi\}}).$$

The Berry–Keating xp Hamiltonian generates a hyperbolic (dilation) flow, and the Selberg trace formula [19] relates the Laplacian spectrum of a hyperbolic surface to the lengths of its closed geodesics — the periodic orbits of the **Anosov geodesic flow**, whose hallmark is exponential mixing (scrambling). The Z-sector scrambling and the geodesic-flow mixing are thus the same kind of dynamics, which is the defensible content of the i-tetration $\leftrightarrow$ hyperbolic-flow correspondence. Two caveats keep this precise: the i-tetration fixed point is an *attractor* ($|\lambda| < 1$), not an Anosov saddle, so the correspondence is through mixing/scrambling, not through an equality of Lyapunov exponents; and $\alpha_{BK} = 0.5664$ is a Z-Spin-specific dilation rate, not the curvature—1 geodesic exponent ($= 1$). [DERIVED-interpretation for the mixing correspondence; the literal exponent identification is OPEN.]

6.1 The $s = 1/2$ confluence

The same Selberg bridge connects the scrambling to a number-theoretic structure. ZS-M4 Theorem 4 (PROVEN) shows the seam-involution mirror-adjointness $JL_s^\dagger J = L_{1-s}$ holds iff $\sigma = 1/2$, marking the critical line as the unique unitarity line of the transfer operator. Independently, Anantharaman and Monk proved in 2025 [10,11] that typical (Weil–Petersson random) hyperbolic surfaces of large genus attain the optimal spectral gap $\lambda_1 \rightarrow 1/4$, where $1/4 = s(1-s)$ at $s = 1/2$ is the bottom of the L^2 -spectrum of $\mathbb{H}^2$ and the tempered (unitary, principal-series) threshold. Both single out the same $s = 1/2$ fixed line of the same $s \leftrightarrow 1-s$ functional-equation symmetry: ZS-M4’s ‘ $\sigma = 1/2$ is the unique unitarity line’ is the operator mirror of ‘typical hyperbolic surfaces are maximally tempered.’ [HYPOTHESIS-strong.]

6.2 Can the confluence be lifted to a theorem? (Test of gate F-PN.1)

A DERIVED status would require an actual operator map from ZS-M4’s L_s to a Selberg/hyperbolic-Laplacian operator. We tested whether such a map exists in the $P_{\max} \rightarrow \infty$ limit. The phase matrices W_p are diagonal, so L_s is a diagonal operator on a **fixed** $\mathbb{C}^{11}$; its eigenvalues are eleven prime-phase Dirichlet sums (the $j = 5$ entry being the prime zeta function). A Selberg/Laplacian operator is, by contrast, infinite-dimensional with a continuous-plus-discrete spectrum and heat-kernel structure. At fixed $Q = 11$ the map is therefore categorically impossible (finite versus infinite dimension); and the

infinite-dimensional (Fock/trace-class, $Q \rightarrow \infty$) extension is exactly the construction ZS-M4 itself flags as conjectural and unbuilt (its O1, O3, and Gap 1 / P1–P4). The lift is thus blocked at the same gap that blocks ZS-M4’s Riemann connection. Gate F-PN.1 does not fully fire — impossibility at $Q \rightarrow \infty$ is not shown, only non-construction — but the conclusion is firm: the confluence cannot presently be raised above HYPOTHESIS-strong. It is not, however, ‘merely numerical’: what the two results share is the $s \leftrightarrow 1-s$ symmetry and its tempered fixed line, which is operator-independent, so the structural status is genuine. [Lift route: OPEN; confluence: HYPOTHESIS-strong.]

Lemma M43.3 (No-Operator-Map at fixed Q). The fixed-Q transfer operator L_s , being diagonal on $\mathbb{C}^Q$ with $Q = 11$, has a spectrum of at most $Q = 11$ points. The Laplacian $-\Delta$ on a closed hyperbolic surface of genus g has area $4\pi(g-1)$ (Gauss–Bonnet) and, by the Weyl law, an eigenvalue counting function $N(\lambda) \sim (g-1)\lambda \rightarrow \infty$ — an infinite discrete spectrum (with, for finite-volume cusped surfaces, an additional continuous part on $[1/4, \infty)$). A similarity or unitary equivalence preserves the cardinality of the spectrum, and no map can take an at-most-11-point spectrum to an infinite one. Hence at fixed Q no correspondence $L_s \rightarrow$ (hyperbolic Laplacian / Selberg operator) exists — a dimension/cardinality obstruction, independent of the detailed entries. It is removed only by $Q \rightarrow \infty$ (an inductive-limit / Fock construction), which is exactly ZS-M4’s conjectural and unbuilt O1/O3/Gap 1. This is the rigorous form of the F-PN.1 verdict. [PROVEN at fixed Q ; OPEN at $Q \rightarrow \infty$.]

Anti-numerology declarations. Two numerical near-coincidences are explicitly declined. (i) The recurring ‘ $1/2$ ’ — $j = 1/2$ (spin invariant, ZS-M3), $\sigma = 1/2$ (critical line), $s = 1/2$ — is claimed as one structural object ONLY for $\sigma = 1/2 \leftrightarrow s = 1/2$; $j = 1/2$ has a distinct origin and is not asserted to be the same. (ii) Embedding α_{BK} as a curvature scale gives a base-of-spectrum $|z^*|^2/4 \approx 0.0805$ (and $\alpha_{\text{BK}}^2/4 \approx 0.0802$) numerically near $A = 0.0801$; this 0.1–0.5% agreement has no derivation and is NOT claimed.

§7. Falsification Gates and Non-Claims

Table 7.1. Falsification gates for ZS-M43.

Gate	Trigger condition
F-M43.1	If a future construction shows the Z-Goldstone does reach a thermal state with a well-defined temperature and thermal entropy, the §3 retirement of η/s is wrong and a thermal hydrodynamics must be built after all.
F-M43.2	If the Goldstone-phase Euler does not reduce to the corpus $\rho_\theta \propto 1/r^2$ in the static curl-free limit, §2 is inconsistent with ZS-A1/A2/F1. (Not triggered.)
F-M43.3	If the scrambling light-cone is measured (or derived) to exceed the Lieb–Robinson velocity, $v_B \leq p(\mathcal{L}) \cdot a$ fails and §5 is wrong (cross-link to ZS-M17 gate F-M17.5, hardware 2027+).
F-M43.4	If the i-tetration leak $ \lambda^2 \approx 0.795$ or the master-equation rates $2A/Q$, $6A/Q$ were shown not to govern X-observer decoherence, the scrambling identification of §4 fails.
F-M43.5	If an operator map $L_s \rightarrow$ Selberg/Laplacian were constructed at $P_{\text{max}} \rightarrow \infty$, the $s = 1/2$ confluence would be promoted to DERIVED (a promotion gate, not a refutation).

Non-Claims

NC-M43.1. No derivation of the Navier–Stokes equations is claimed; Route B yields a coherent Euler/superfluid plus a scrambling structure, not a thermal Navier–Stokes.

NC-M43.2. No thermal transport coefficient, no thermal entropy density, and no η/s value is claimed; these are declared inapplicable, not merely unknown.

NC-M43.3. The Euler limit and the OTOC/butterfly framework are standard physics; the Z-Spin-specific content is the parameter-free scrambling structure ($|\lambda^2|$ leak; $2A/Q$, $6A/Q$) and its $v_B \leq \rho(\mathcal{L}) \cdot a$ bound.

NC-M43.4. The $s = 1/2$ confluence is a structural correspondence, NOT an operator identity and NOT a Riemann-Hypothesis claim; ZS-M4's own non-claims stand.

§8. Conclusion

Route B of ZS-M42 is resolved. The Z-Goldstone is a coherent Euler/superfluid field with the corpus halo $\rho_\theta \propto 1/r^2$ as its static instance, and it has no thermal viscosity: η/s and the KSS bound are inapplicable because the Goldstone never thermalizes and the dynamics is unitary. The dissipation that does exist is unitary scrambling, a parameter-free structure governed by the i-tetration leak $1-|\lambda^2| \approx 0.205$ per cycle and the master-equation rates $\lambda_{\text{fast}} = -A$, $\gamma_{xz} = 2A/Q$, $\gamma_{zy} = 6A/Q$, propagating at a butterfly velocity $v_B \leq \rho(\mathcal{L}) \cdot a$. Its geometric model is Anosov geodesic mixing; via the Selberg bridge it meets ZS-M4's critical line at the $s = 1/2 \leftrightarrow \lambda = 1/4$ confluence, which a direct operator map cannot presently raise above HYPOTHESIS-strong. The methodological lesson is general for the corpus: when an estimate in a Z-Spin sector is unstable under re-derivation, the productive move is to ask whether the imported framework (here thermal hydrodynamics) matches the corpus ontology (here unitary scrambling), rather than to refine a coefficient.

Acknowledgements & Code Availability

This paper consolidates the June 2026 ZS-M42 Route-B program, including the dependency audit against the ZS-M17 dated-erratum (2026-06-07). The numerical checks (the viscosity-instability and temperature comparisons of Appendix C, the $|\lambda^2|$ leak, and the structure of the ZS-M4 transfer operator) are elementary and reproducible from the locked constants $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$, the i-tetration fixed point z^* (ZS-M1), and $\xi \approx 0.75 \ell_P$ (ZS-Q5); a short verification script is available on request.

Appendix A. Structural-Consistency Checklist (22/22 PASS)

Table A.1. Structural-consistency checks for ZS-M43 v1.4.

#	Check	Verdict
C1	Zero new free parameters; no scale beyond A , Q , $\dim(Z) = 2$, z^* , ξ , τ_P .	PASS
C2	Goldstone θ massless, $Q = a^3 \rho^2 \theta$ conserved (ZS-F1 / ZS-U11).	PASS
C3	Relativistic Euler from $T_{\mu\nu}$ conservation — standard superfluid hydro (NC-M43.3).	PASS
C4	Static curl-free limit matches corpus $\rho_\theta \propto 1/r^2$ (ZS-A1/A2/F1).	PASS
C5	θ never thermalizes (ZS-F1 §7.2); dynamics unitary, entropy = scrambling (§12.1).	PASS
C6	Hence thermal T , s , η/s , and the KSS bound are inapplicable (§3; Appendix C instability).	PASS
C7	Scrambling rotation $\arg \lambda \approx 129.4^\circ/\text{step}$ and leak $ \lambda^2 \approx 0.795$ cited from ZS-M1 (PROVEN).	PASS

#	Check	Verdict
C8	Coarse-grained rates $\lambda_{\text{fast}} = -A$, $\gamma_{xz} = 2A/Q$, $\gamma_{zy} = 6A/Q$ from ZS-Q7/F0 (PROVEN/DERIVED).	PASS
C9	Butterfly velocity $v_B \leq v_{LR} = \rho(\mathcal{L}) \cdot a$ (ZS-M17, $\leq$ -form per erratum); generically strict.	PASS
C10	MSS chaos bound and KSS bound correctly noted inapplicable (non-thermal) — no thermal-bound overreach.	PASS
C11	$\alpha_{BK} = -\ln z^* = 0.5664$ cited from ZS-M4 Theorem 3 (PROVEN); i-tetration is an attractor, not Anosov.	PASS
C12	$s = 1/2 \leftrightarrow \lambda = 1/4$ confluence: same $s \leftrightarrow 1-s$ symmetry; HYPOTHESIS-strong, not numerical.	PASS
C13	F-PN.1 tested: L_s diagonal on fixed $\mathbb{C}^{11}$; map to Laplacian impossible at fixed Q , OPEN at $Q \rightarrow \infty$.	PASS
C14	Anti-numerology: $ z^* ^2/4 \approx 0.0805 \approx A$ and the recurring $1/2$ are declined, not claimed (§6.2).	PASS
C15	EOS/sound speed (§2.1): kination $w = 1$, $c_s = 1$ (stiff/luminal); static halo $p_r = \rho$, $p_t = -\rho$, $\bar{w} = -1/3$, $\rho \propto 1/r^2$ — matches corpus.	PASS
C16	Dynamical character (§4.1): $KS = 0$ (attracting, non-chaotic, $\lambda_L = 0$); fixed-rate decoherence $1/A$; $t_* \sim \ln S$ explicitly inapplicable (§5).	PASS
C17	Lemma M43.1 (§3): unitary + non-thermalizing $\Rightarrow$ Kubo η and thermal s undefined; $\eta/s = 0/0$ ($T \rightarrow 0$ normal component); KSS vacuous. DERIVED.	PASS
C18	Prop. M43.2 (§4.1): $\dot{S}_X \cdot T_{\text{cycle}} = -\ln \lambda^2 = 2(\Gamma_Z \cdot T_{\text{cycle}})_{\{F16\}}$ (0.16%); Markovian $2/Q$ vs 0.205 ($\sim 11\%$); $\leq \ln 2$ (Q7). Consistent.	PASS
C19	§2.2: kination $\rho_\theta = Q_w^2/(2a^6\epsilon^2) \propto a^{-6} = \text{ZS-M12 centrifugal term}$, $Q_w = A$; near-bounce phase; $w=1$ anisotropy-marginal + Q -survival OPEN (honest).	PASS
C20	Remark to Lemma M43.1 (§3): two-fluid $\eta \propto \rho_n$, $s \propto \rho_n$; $T=0$ condensate $\rho_n \neq 0 \Rightarrow \eta=s=0 \Rightarrow 0/0$; KSS (finite- T , normal-dominated) out of domain. Landau [28].	PASS
C21	§2.2 (quantified): $\rho_\theta/\rho_{\text{shear}} = A^2/(2\epsilon^2 C_\sigma) \text{ const (both } \propto a^{-6})$; $A^2 \approx 6.4e-3 \Rightarrow$ anisotropy-subdominant; complements thermal $\tau_{\text{thermal}}/\tau_{AS} \approx 0.81$; Auto-Surgery necessary.	PASS
C22	Lemma M43.3 (§6.2): fixed- Q diagonal L_s has ≤ 11 -pt spectrum; hyperbolic Δ infinite (Weyl) + cont $[1/4, \infty)$; cardinality obstruction $\Rightarrow$ no map. PROVEN at fixed Q ; OPEN at $Q \rightarrow \infty$.	PASS

Appendix B. Cross-Paper Dependency and Version-Conflict Audit

All imported corpus results are used as stated; none is modified, and the constants A , Q , $\dim(Z) = 2$ are used identically throughout. No version conflict.

Table B.1. Upstream dependencies and status.

Source	Result used	Status / conflict check
ZS-M42 v1.2	Two-Gate No-Go; Route B mandate	Consistent (this paper closes Route B)
ZS-M17 (erratum 2026-06-07)	$v_{\text{max}} \leq \rho(\mathcal{L}) \cdot a$ ($\leq$ -form)	Consistent; $\leq$ used for v_B bound, not the retracted equality
ZS-F1 §7.2 / §12.1	θ non-thermalizing; unitary scrambling	Load-bearing (§3, §4); used as stated
ZS-A1 / A2 / F1	$\rho_\theta \propto 1/r^2$ (DERIVED)	Static instance (§2)
ZS-M1	z^* , $ \lambda = 0.8916$, $ \lambda^2 \approx 0.795$ (PROVEN)	Scrambling rotation/leak (§4)

Source	Result used	Status / conflict check
ZS-Q7 / ZS-F0	$\lambda_{\text{fast}} = -A, \gamma_{xz} = 2A/Q, \gamma_{zy} = 6A/Q$	Coarse-grained rates (§4)
ZS-M4 (Thms 3, 4)	$\alpha_{\text{BK}} = -\ln z^* $; J-mirror iff $\sigma = 1/2$	Geometric model and confluence (§6)
KSS 2005; MSS 2016	$\eta/s \geq 1/4\pi; \lambda_L \leq 2\pi T$ (external)	Used as physicality filters; shown inapplicable (non-thermal)
Anantharaman–Monk 2025	$\lambda_1 \rightarrow 1/4$ optimal gap (external)	Geometric anchor of the confluence (§6.1)

Appendix C. The Viscosity Instability, for the Record

This appendix documents the dead-end that §3 summarizes, so that it is not re-attempted. If one nonetheless treats the Goldstone drag ($O(A^2/M_P^2)$ per oscillation, ZS-U11 NC-U11.1 [6]) as a transport input and writes $\eta/s = \tau_R \cdot T$ (with $\eta = (\rho+p)\tau_R$, $s = (\rho+p)/T$ at $\mu = 0$), the result depends entirely on two unfixed choices, as Table C.1 shows.

Table C.1. The two instabilities that signal a wrong framework.

Choice	Outcome	Verdict
Drag as mode-damping $\rightarrow v \sim A^2 \xi^2 \approx 0.0036 \ell_P$	$\eta/s \approx 4 \times 10^{-4} \ll 1/4\pi$	KSS-violating $\rightarrow$ unphysical
Drag as relaxation rate $\rightarrow v \sim \tau_P/A^2 \approx 156 \ell_P$	$\eta/s \sim O(10^{-10^2})$	Opposite scaling; no unique value
Temperature: cyclic vs core	$T \approx 0.11\text{--}0.36$ vs $0.21 M_P$ ($\times \sim 2\text{--}3$)	Not unique $\rightarrow \eta/s$ not parameter-free

The sign of the A -exponent is undetermined and the temperature is non-unique; by the convergence test of the deep-exploration protocol, this instability is the diagnostic that the thermal-fluid framework does not fit, which §3–§4 act on. The KSS bound is used here only as a physicality filter; it is not a claim about the Z -fluid.

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Version History

v1.4 (June 2026): Second-layer deepening of the same three sections (no new fragment). The three v1.3 claims are tightened with rigorous limiting arguments. (§3) A two-fluid Remark makes Lemma M43.1(c) sharp: η and s both reside in the Landau normal component $\rho_n(T)$, which vanishes as $T \rightarrow 0$; the non-thermalizing condensate sits at $\rho_n \equiv 0$, so $\eta = s = 0$ and $\eta/s = 0/0$, and the KSS bound (a normal-component-dominated, finite- T statement) is out of domain. (§2.2) The $w = 1$ anisotropy-marginality is quantified: ρ_θ and the Bianchi shear both $\propto a^{-6}$, so their ratio is a constant $\propto A^2 \approx 6.4 \times 10^{-3}$, making the Goldstone anisotropy-subdominant — the deterministic complement to the corpus thermal-shear constraint $\tau_{\text{thermal}}/\tau_{\text{AS}} \approx 0.81$ — so the topological Auto-Surgery, not stiff dilution, must smooth the bounce. (§6.2) Lemma M43.3 (No-Operator-Map) gives the rigorous form of F-PN.1: the fixed- Q diagonal L_s has an at-most-11-point spectrum while the hyperbolic Laplacian has an infinite (Weyl) spectrum, a cardinality obstruction that no similarity can bridge; the lift is OPEN only at $Q \rightarrow \infty$. Structural checks 19 $\rightarrow$ 22; references +2 ([28–29]). Zero new free parameters; $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$ unchanged.

v1.3 (June 2026): Internal-density deepening (no new fragment). Three previously-prose points are made into formal, cross-validated claims. (§3) Lemma M43.1 (No-Thermal-Transport) states that global unitarity plus non-thermalization make the Kubo η and the thermal entropy density undefined, so η/s is the indeterminate $0/0$ in the pure-coherent ($T \rightarrow 0$) limit and the KSS bound is vacuous. (§4.1) Proposition M43.2 cross-validates the X-observer entropy-production rate $\dot{S}_X$ three ways: exactly against the ZS-F16 Wilson dissipation ($\dot{S}_X \cdot T_{\text{cycle}} = -\ln|\lambda^2| = 2 \cdot \Gamma_Z \cdot T_{\text{cycle}}$, 0.16%), approximately against the ZS-Q7 Markovian rates ($2/Q$ vs $1-|\lambda^2|$, $\sim 11\%$ expected error), and as bounded by the Z-channel capacity $\ln 2$. (§2.2) The stiff $w = 1$ regime is identified with the ZS-M12 centrifugal-launch term $\rho_\theta = Q_w^{-2}/(2a^6\epsilon^2) \propto a^{-6}$ with comoving winding $Q_w = A$, placing it at the near-bounce phase of the ZS-U11 cycle; the $w = 1$ anisotropy-marginality and the Q -survival through the $\approx 3\tau_P$

Auto-Surgery are flagged OPEN. Structural checks 16 $\rightarrow$ 19; references +3 ([25–27]). Zero new free parameters; $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$ unchanged.

v1.2 (June 2026): Mathematical-density deepening of the existing sections (no new fragment, no new scope). §2.1 adds the equation of state and sound speed, derived with zero free parameters — a stiff fluid ($w = 1$, $c_s = 1$) in the time-dependent regime and an anisotropic global-monopole configuration ($p_r = \rho$, $p_t = -\rho$, $\bar{w} = -1/3$, $\rho \propto 1/r^2$) in the static-halo regime, matching the corpus halo. §4.1 pins the dynamical character of the scrambling: the i-tetration fixed point is attracting, so by Pesin’s identity the Kolmogorov–Sinai entropy is zero, there is no positive quantum Lyapunov exponent, and the scrambling is unitary information delocalization (CPTP) with a fixed, size-independent decoherence time $1/A$; the correct replacement for η/s is the X-observer entropy-production rate. §5 is sharpened accordingly: the OTOC framework gives a bounded butterfly velocity but no exponential growth, and the fast-scrambling time $t_* \sim \ln S$ is shown explicitly NOT to apply (the Z-sector is a fixed-rate decoherer, not a black-hole-like fast scrambler) — the OTOC counterpart of the MSS-bound inapplicability. Structural checks 14 $\rightarrow$ 16, all PASS; references +4 ([21–24]). Zero new free parameters; $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$ unchanged.

v1.1 (June 2026): Full revision. The paper is reorganized around its constructive result — the Z-sector dissipation is unitary scrambling — with the thermal- η/s retirement reduced to its essential argument (§3) and the unstable viscosity estimate moved to Appendix C as a documented dead-end. New material folded in: §4–§5 elevate scrambling to a first-class subject (i-tetration $|\lambda^2|$ leak; ZS-Q7 master-equation rates; OTOC and the butterfly velocity $v_B \leq \rho(\mathcal{L}) \cdot a$; the non-applicability of the MSS chaos bound), and §6 gives the Anosov / Selberg geometric model together with the $s = 1/2 \leftrightarrow \lambda = 1/4$ confluence and a test of gate F-PN.1 (the operator map $L_s \rightarrow$ Selberg/Laplacian is blocked, so the confluence stays HYPOTHESIS-strong). The standalone position note PN-M4.1 is superseded by §6. Structural checks 12 $\rightarrow$ 14, all PASS. Zero new free parameters; $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$ unchanged.

v1.0 (June 2026): First public release. Established the coherent Euler/superfluid result and the retirement of the thermal η/s programme. (Consolidated from internal Z-Spin Collaboration deep-exploration notes up to v0.4, June 2026.)